

Table 6. The prompt for entity extraction, namely τ_2 .

System Message	Now you are an expert in extracting key entities from research contents. You are good at identifying the most important keywords or phrases that summarize the main topics or concepts discussed in the content.
User Message	Task Description: I will provide you with a content from a research paper. Your task is to extract the key entities from this content. These entities are the most important keywords or phrases that summarize the main topics or concepts discussed in the content. Instruction: Content: The content is your key focus, and the extracted entities should be based on the content. In other words, the entities you extract should be concrete manifestations of the main themes and topics discussed in the content. Your approach should be systematic: - Start by thoroughly reading the content to understand its main themes and topics. - Identify and list the key entities that are central to the content. - Ensure that the entities are relevant, meaningful, and representative of the content. - Each entity in entities should be no longer than 5 words. - Each entity in entities should contain at least 2 words. - The number of entities should be less than or equal to 5. - Each entity in entities should be nouns or noun phrases. examples: {examples} Your turn: Given the following content: {content} Your answer should follow this format: entity1, entity2, entity3,

Table 7. The prompt for extracting concise methods, namely τ_3 .

System Message	Now you are a researcher in the field of AI with innovative and pioneering abilities.
User Message	# Task Description: You are an AI researcher conducting studies in a specific domain. Someone has provided you with a methodology section, and your task is to transform it into another style. I will give you an example. The example begins with “# Example 1” and includes a Example Summarized Methods. Then, your task starts with “# Your Task”, containing “Your Methodology Section”. Your job is to transform Your Methodology Section into a Summarized Methods by referring to Example 1. Note that the ideas in Example 1 are unrelated to your idea, so the key focus should be on the style of Example Summarized Methods. You should directly start with your response and do not start with a section title like “## Your Summarized Methods”. # Example 1 ## Example Summarized Methods {Example Summarized Methods} # Your Task ## Your Methodology Section {methodology} ## Your Summarized Methods

Table 8. The prompt for background transformation.

System Message	You are a teacher in the field of AI, skilled at clearly explaining AI concepts to students. Your student is an undergraduate in AI with a basic understanding of deep learning.
User Message	# Task Description: You are teaching your undergraduate about a specific subfield of AI research. You have a brief description of the research background, and now you need to explain its meaning and purpose in detail to your undergraduate. Keep in mind that your undergraduate may be completely unfamiliar with the technical terms in the research background. I will give you an example. The example begins with “# Example 1” and includes a Brief Research Background, several Technical Terms, and the corresponding Detailed Research Background. Then, your task starts with “# Your Task”, containing “Your Brief Research Background” and “Your Technical Terms”. Your job is to expand Your Brief Research Background into a Detailed Research Background by referring to Example 1. Note that the research background in Example 1 are unrelated to yours, so the key focus should be on the relationship between the Brief Research Background and the Detailed Research Background. You should directly start with your response and do not start with a section title like “## Detailed Background”. # Example 1 ## Brief Research Background {Example user’s query} ## Technical Terms large language models, kv cache, gpu memory ## Detailed Research Background {Example detailed research background} # Your Task ## Your Research Background {brief_background} ## Your Technical Terms {keywords}

Table 9. The prompt for brainstorming.

System Message	Now you are a researcher in the field of AI with innovative and pioneering abilities. You are good at generating creative and original ideas.
User Message	### Task Description: You are an AI researcher tasked with brainstorming initial, innovative ideas to address a given research problem in AI. Focus on generating diverse and creative approaches rather than finalized methods. The ideas can be rough and in their infancy but should cover a range of possible directions that could be explored further. ### Information Provided: - Research Background: {background} ### Approach: Your brainstorming should be systematic: - Step 1: Thoroughly understand the research background. - Step 2: Generate a list of 4 to 6 high-level ideas or directions that could potentially solve problems in the given background. Be creative, think outside the box, and avoid merely rephrasing existing methods. ### Format for Your Response: Please present 4 to 6 ideas in the following format: Idea 1: [Brief description of the first idea] Idea 2: [Brief description of the second idea] ...

Table 10. The prompt for initial idea generation.

System Message	Now you are a researcher in the field of AI with innovative and pioneering abilities. You are good at using innovative and original methods to solve cutting-edge problems in the field of AI.
User Message	# Task Description: You will be provided with a research problem, along with inspirations extracted from related papers. Your task is to identify and combine these inspirations to propose 3 to 4 different ideas to solve the research problem. The ideas should be innovative and try to avoid using the same inspirations repeatedly. Each idea should start with Idea. # Information Provided: ## Research problem {Transformed background} ## Inspirations {Inspirations}